Exp 6a – Molecular Modeling

INTRODUCTION

Lewis structures and ball-and-stick models are different ways to represent molecules and polyatomic ions. Lewis structures show how atoms are connected and arranged in molecules and polyatomic ions (the bonds and lone pairs) but since Lewis structures are drawn on a flat piece of paper, they do not accurately show the shape of molecules and polyatomic ions. The shape (three-dimensional representations) of molecules determines their physical and chemical properties. Ball-and-stick models are three-dimensional representations of molecules and polyatomic ions. In this experiment you will build models for molecules and draw their Lewis structures.

BACKGROUND

The model kit consists of colored balls and gray connectors. The shorter, rigid connectors (5/kit) are used to represent single bonds between two atoms. The longer, flexible connectors (4/kit) can represent double and triple bonds. Two connectors are used for a double bond (Figure 1) and three are used for a triple bond. The model kit uses different colors to represent different atoms (see table 1 below).

Table 1: Model Kit Composition

Color	*Element*	*# holes*	*#/kit*
white	hydrogen	1	4
black	carbon	4	3
red	oxygen	2	2
green	chlorine	1	2
orange	bromine	1	2
purple	iodine	1	2
blue	nitrogen *	3	2



Figure 1. Models showing a single bond (left) and a double bond (right)

*

Many blue nitrogen balls have four holes, but one hole is plugged with a small white peg, glue, or covered with tape so that it only has three available holes. All three holes in each ball will have a connector when the model is built correctly.

Lewis structures use dots to represent the valence electrons, but these drawings are two dimensional and often do not accurately depict the three-dimensional shape of molecules. The molecular modeling sets help visualize the three-dimensional shape of the molecules. Valence Shell Electron Pair Repulsion

Theory allows us to assign shape names to the observed three-dimensional shapes. For example, the Lewis structure for OCl_2 drawn in Figure 2 shows the atoms connected in a straight line, which does not accurately represent the shape of the molecule. The central oxygen has four electron groups around it (Figure 2). A molecule with four electron groups around the central atom has a ***tetrahedral*** electron group geometry because those electron groups want to get as far apart as possible. Two of the electron groups are lone pairs. Although the lone pairs affect the three-dimensional shape, they cannot be seen. The observed molecular geometry is ***bent***. The names for all the electron group geometries and molecular geometries can be found in Table 1.

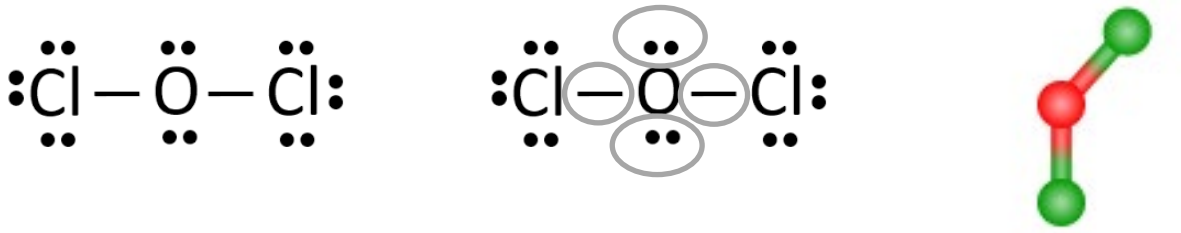


Figure 2: A) The Lewis structure for OCl_2 drawn where it appears to be linear. B) The Lewis structure for OCl_2 with electron groups circled. C) The molecular model for OCl_2 shows bent molecular geometry.

Number of Electron Groups	Electron Group Geometry	Ideal Bond Angles	0 lone pairs	1 lone pair	2 lone pairs
			Molecular Geometry		
2	Linear	180°	Linear		
3	Trigonal Planar	120°	Trigonal Planar	Bent	
4	Tetrahedral	109.5°	Tetrahedral	Trigonal Pyramidal	Bent

Table 1. Electron group geometry and molecular geometry are based on the number of electron groups.

ADDITIONAL READING

- Lewis Structures, VSEPR

SAFETY EQUIPMENT REQUIRED FOR LAB

- No safety equipment needed

SUPPLY LIST

- Molecular model kit
- Color Pencils

PROCEDURE – PART 1 – LEWIS STRUCTURE FIRST

1. ☐ Using the periodic table, find the total number of valence electrons based on the chemical formula.
2. ☐ Sketch the Lewis Dot structure.
    a. Each atom (except hydrogen) should have an octet (8 electrons). Hydrogen should have a duet (2 electrons).
3. ☐ Build a molecular model based on your Lewis Dot Structure. Each hole on the model will be filled when the model is correctly assembled.
    a. Use short grey connectors to make single bonds, filling one hole in each atom.
    b. Use long grey connectors to make double or triple bonds. A double bond requires two long grey connectors between two atoms. A triple bond requires three connectors between two atoms.
4. ☐ Sketch the molecular model in the *Molecular Model* column showing the three-dimensional structure.
    a. Show single bonds with single lines, double bonds with two lines and triple bonds with three lines.
5. ☐ Use colored pencils to color your drawing.
6. ☐ Count the electron groups around the central atom (or atom specified). Use that information to determine the electron group geometry and molecular geometry around that atom.

PROCEDURE – PART 2 – MOLECULAR MODEL FIRST

7. ☐ Similar to before, only now attempt to build the model BEFORE drawing the Lewis Structure. You will know your model is correct when each hole on the model is filled.
8. ☐ Your instructor will provide models of unknown molecules for the last part. For each molecule:
    a. Sketch the model using the appropriate colors.
    b. Write the chemical formula (in the first column).
    c. Determine the number of valence electrons.
    d. Draw the Lewis structure.

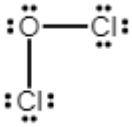
DATA AND REPORT – DUE ____ / ____ / ____ **SECTION** _____

* Show your work. **DATE** _____

PARTNER _____

Part 1: Draw the Lewis Structure First

MOLECULAR MODELS WITH SINGLE BONDS ONLY

#0	OCl_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
20			4 **Electron Group Geometry:** Tetrahedral **Molecular Geometry:** Bent

#1	H_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			 Electron Group Geometry: linear **Molecular Geometry:** linear

#2	HCl		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			 Electron Group Geometry: linear **Molecular Geometry:** linear

#3	**CH_2Cl_2**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			Electron Group Geometry:
			Molecular Geometry:

#4	**H_2O**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			Electron Group Geometry:
			Molecular Geometry:

#5	**NH_3**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			Electron Group Geometry:
			Molecular Geometry:

MOLECULAR MODELS ONE DOUBLE BOND

#6	**O_2**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			~~X~~
			Electron Group Geometry:
			Trigonal planar
			Molecular Geometry:
			linear

#7	**C_2H_4**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

#8	**C_2H_3Cl**		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

MOLECULAR MODELS ONE TRIPLE BOND

#9	N_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
			Electron Group Geometry: linear
			Molecular Geometry: linear

#10	C_2H_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

Part 2: Build the Model First

MOLECULAR MODELS WITH SINGLE BONDS ONLY

#11	H_2O_2			
Molecular Model Sketch		**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around O:**
				Electron Group Geometry around O:
				Molecular Geometry around O:

#12	N_2H_4			
Molecular Model Sketch		**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around N:**
				Electron Group Geometry around N:
				Molecular Geometry around N:

#13	NH_2OH			
Molecular Model Sketch		**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around N:**
				Electron Group Geometry around N:
				Molecular Geometry around N:

MOLECULAR MODELS WITH ONE DOUBLE BOND

#14	**HONO**		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around N:**
			Electron Group Geometry around N:
			Molecular Geometry around N:

#15	**HCOOH**		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

MOLECULAR MODELS WITH ONE TRIPLE BOND

#16	**HOCN**		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

MOLECULAR MODELS WITH TWO DOUBLE BONDS

#17	CO_2		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around C:**
			Electron Group Geometry around C:
			Molecular Geometry around C:

#18	C_3H_4		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around central C:**
			Electron Group Geometry around central C:
			Molecular Geometry around central C:

UNKNOWN MOLECULAR MODELS

Molecule	Molecular Model	Valence Electrons	Lewis Structure
#1 Chemical Formula: _____			
#2 Chemical Formula: _____			
#3 Chemical Formula: _____			
#4 Chemical Formula: _____			
#5 Chemical Formula: _____			